DaVita ADT Alert–Triggered SMS Outreach & AI Care Summary

Post-Discharge Patient Outreach and Care Team Notification

1. Executive Summary

DaVita care teams often do not receive timely information when a patient is admitted, discharged, or transferred outside the clinic. Although ADT (Admission, Discharge, Transfer) alerts are available through external HIE/Interop feeds, the follow-up process is still largely manual.

As a result:

- Care teams may not know a patient was recently hospitalized.
- Important discharge information may be missed.
- Medication changes may not be captured quickly.
- Symptoms after discharge may go unnoticed.
- Follow-up actions may be delayed.

This POC demonstrates an AI-assisted workflow where:

1. An ADT discharge alert is received.
2. The system automatically sends an SMS survey to the patient.
3. The patient responds in natural language.
4. AI extracts useful clinical insights.
5. A structured summary is generated.
6. The care team receives actionable information.

The goal is to improve post-discharge visibility, reduce manual outreach, and support faster care coordination.

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2. Problem Statement

Current Challenge

DaVita depends on external HIE/interoperability systems to receive ADT alerts when patients are hospitalized.

However:

- Alerts may not be reviewed immediately.
- Follow-up workflows are manual.
- Care teams often lack discharge context.
- Patient symptoms after discharge are not proactively collected.
- Medication updates are delayed.
- High-risk patients may not be escalated quickly.

Business Impact

Without timely outreach:

- Patients may deteriorate after discharge.
- Medication reconciliation may be delayed.
- Care coordination gaps increase.
- Risk of readmission increases.
- Care teams spend additional manual effort collecting information.

Proposed Solution

Build an automated workflow where:

- ADT discharge alerts trigger SMS outreach.
- Patients answer simple post-discharge questions.
- AI interprets free-text responses.
- Rule-based logic identifies risk.
- AI creates a concise care summary.
- Care teams receive actionable insights.

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3. Objectives of the POC

Primary Objectives

- Demonstrate ADT-triggered outreach.
- Automate patient follow-up after discharge.
- Use AI to summarize patient responses.
- Identify symptoms and medication changes.
- Deliver structured summaries to care teams.

Secondary Objectives

- Reduce manual outreach workload.
- Improve visibility into post-discharge patient condition.
- Enable earlier nurse intervention.
- Validate AI-assisted workflow feasibility.

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4. Scope of the POC

- Mock/sample ADT event generation
- Event listener service
- Discharge event validation
- SMS survey workflow
- Patient response capture
- AI extraction and summarization
- Rule-based risk detection
- Care-team dashboard or DSM notification

5. Why Use Mock JSON Instead of Real HL7/FHIR?

Real healthcare integrations are complex and time-consuming.

HL7/FHIR integration requires:

- Interface engines
- Vendor coordination
- Security approvals
- Mapping and transformation logic
- Environment setup
- Testing cycles

For a POC, the goal is to validate the workflow and AI capabilities quickly.

Therefore, mock JSON ADT events are used to:

- Simulate discharge alerts
- Trigger the workflow
- Reduce integration dependency
- Accelerate development
- Focus on AI and automation

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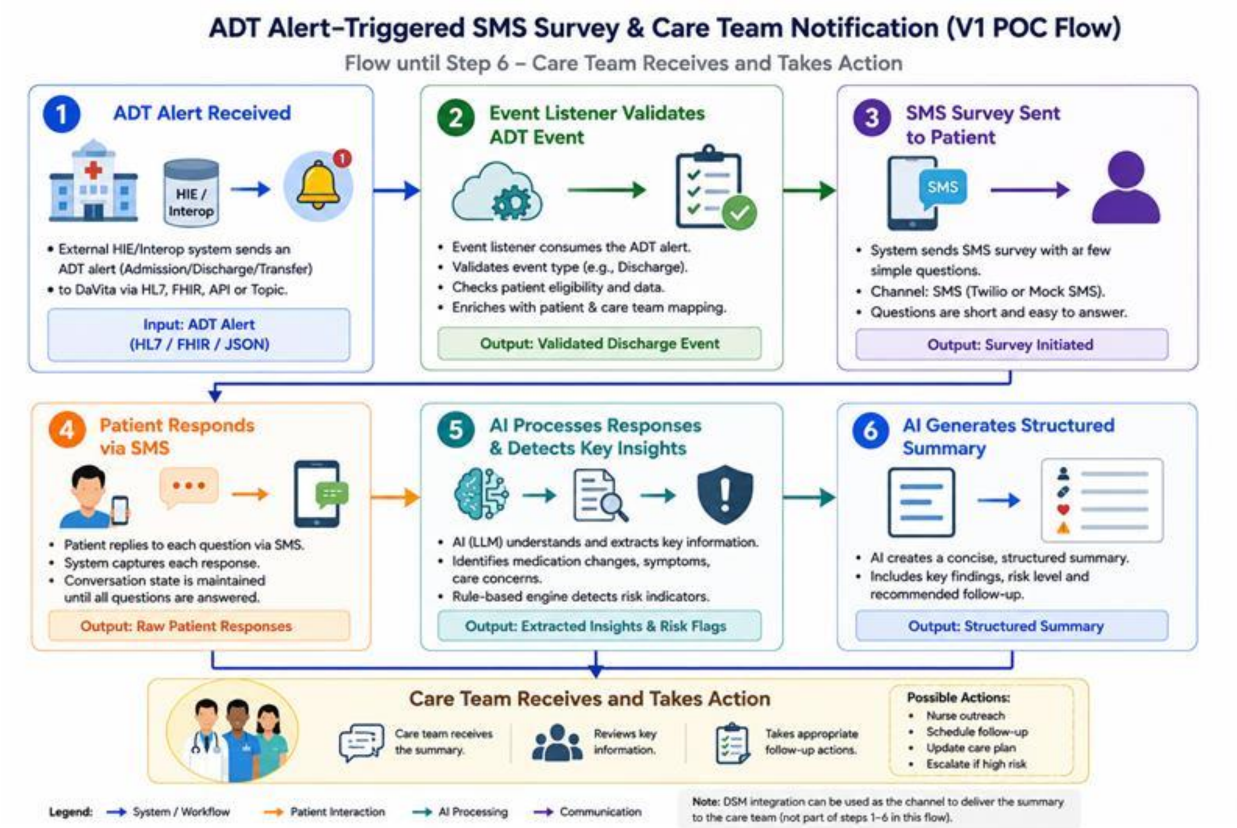
6. High-Level Workflow



End-to-End Flow

1. Hospital generates ADT event.
2. HIE/Interop sends ADT alert.
3. Backend service receives event.
4. System validates discharge event.
5. SMS survey is sent to patient.
6. Patient replies via SMS.
7. AI processes responses.
8. Risk rules evaluate symptoms.
9. AI generates structured summary.
10. Care team receives summary.
11. Care team takes follow-up action.

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7. Detailed Workflow Explanation

Step 1 – ADT Alert Received

Description

A hospital event occurs:

- Admission
- Discharge
- Transfer

The HIE/interoperability feed sends an ADT alert.

Example

Patient X discharged today.

Purpose

This ADT event acts as the trigger for the entire workflow.

Example Mock ADT JSON


```json
{
  "eventType": "DISCHARGE",
  "patientId": "12345",
  "patientName": "John Doe",
  "hospital": "General Hospital",
  "dischargeDate": "2026-05-12",
  "phone": "+15555551234"
}
```


Inputs

- Mock JSON ADT event
- HL7/FHIR event (future state)

Output

- Validated incoming event

Step 2 – Event Listener Validates ADT Event

Description

A backend listener receives the ADT event and validates:

- Is this a discharge event?

- Is patient eligible?
- Does patient have SMS consent?
- Should outreach occur?

Example Events

Event Type	Action
Admission	Ignore for V1
Transfer	Ignore for V1
Discharge	Trigger SMS survey

Output

Validated discharge event.

Step 3 – SMS Survey Sent to Patient

Description

After validation, the system sends SMS questions.

Example Questions

1. Were you discharged from the hospital?
2. Were any new medications prescribed?
3. Are you experiencing any symptoms now?
4. Anything your care team should know?

Example SMS Conversation

```
System: Were you discharged from the hospital?
Patient: Yes

System: Were any new medications prescribed?
Patient: Yes, blood pressure medication.

System: Are you having any symptoms now?
Patient: Swelling in my legs.
```

Output

- SMS survey delivered

Step 4 – Patient Responds via SMS

Description

The patient replies in natural language.

Example Responses

```
Yes, I was discharged yesterday.
Doctor added a new BP medication.
I still have swelling in my legs.
```

System Responsibilities

- Capture responses
- Store message history
- Associate responses with patient/event
- Prepare data for AI processing

Output

Raw patient responses.

Step 5 – AI Processes Responses

Description

AI analyzes the patient's free-text responses and extracts clinical insights.

Example Input

```
Doctor added a new BP medication and I still have swelling.
```

AI Extracts

Information Type	Extracted Value
Medication Change	Yes
Symptom	Swelling
Concern	Possible fluid issue

Why AI Is Valuable Here

Patients respond using natural language.

AI helps:

- Understand intent
- Extract symptoms

- Detect medication changes
- Identify concerns
- Summarize context

Why Rules Are Still Important

Risk classification should remain rule-based for safety.

Example Risk Rules

Symptom	Risk Level
Chest pain	High
Shortness of breath	High
Severe dizziness	High
Swelling	Medium
No symptoms	Low

Example AI Output

```json
{
  "medicationChange": true,
  "symptoms": ["leg swelling"],
  "riskLevel": "Medium",
  "recommendedAction": "Nurse follow-up"
}
```

Output

Extracted insights and risk flags.

Step 6 – AI Generates Structured Summary

Description

AI converts the patient interaction into a concise care-team summary.

Example Summary

```
Patient confirmed recent hospital discharge.
New blood pressure medication prescribed.
Current symptom reported: leg swelling.
Risk Level: Medium.
Recommended Action: Nurse follow-up.
```

Goals of the Summary

- Easy to read

- Actionable
- Clinically useful
- Consistent structure
- Fast review by care teams

Output

Structured care summary.

8. Proposed Architecture

High-Level Components

1. ADT Event Generator

Simulates incoming ADT alerts.

2. Event Listener Service

Receives and validates events.

3. Rules Engine

Determines whether outreach should occur.

4. SMS Service

Sends and receives SMS messages.

5. Response Storage

Stores patient responses and survey state.

6. AI Processing Service

Extracts insights and generates summaries.

7. Risk Detection Engine

Applies deterministic safety rules.

8. Care Team Dashboard

Displays summaries and risk indicators.

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9. Suggested Technical Architecture

Frontend

- React dashboard
- Demo SMS UI
- Care-team summary view

Backend

- Python Fast API
- Event processing service

AI Layer

- OpenAI GPT model
- Prompt-based extraction
- Structured JSON generation

Messaging

- Twilio SMS API
- Mock SMS simulator

Database

- PostgreSQL
- SQLite for demo

10. Data Flow

Input

Mock ADT JSON event.

Processing

1. Event validation
2. SMS orchestration
3. Response collection
4. AI extraction
5. Risk scoring
6. Summary generation

Output

Structured care-team summary.

11. Required Components for the POC

Infrastructure Requirements

- Backend API service
- SMS provider account
- AI API access
- Database
- Dashboard UI

Functional Requirements

- Receive ADT events
- Validate discharge events
- Send SMS survey
- Capture responses
- Process responses using AI
- Apply risk rules
- Generate summary
- Display care-team view

12. Suggested APIs

ADT Event API

```
POST /api/adt-event
```

Send SMS API

```
POST /api/send-survey
```

Receive SMS Webhook

```
POST /api/sms-response
```

AI Processing API

```
POST /api/process-response
```

Summary API

```
GET /api/patient-summary/{patientId}
```

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13. Example End-to-End Scenario

Scenario

1. Hospital discharges patient.
2. Mock ADT event sent.
3. System validates discharge.
4. SMS survey sent.
5. Patient reports:
    - New BP medication
    - Leg swelling
6. AI extracts insights.
7. Rules classify risk as Medium.
8. Summary generated.
9. Care team members receives dashboard alert.

14. Conclusion

This POC demonstrates how ADT alerts, SMS outreach, AI processing, and structured summaries can improve post-discharge patient engagement and care-team visibility.

The design intentionally separates:

- Rules-based workflows for safety and predictability
- AI-based workflows for natural language understanding and summarization

Using mock ADT events allows rapid validation of the workflow without the complexity of real HL7/FHIR integration.

The solution provides a strong foundation for future production-ready interoperability and AI-assisted care coordination workflows.